QKD Simulator Survey

Quantum Key Distribution as an Educational Tool

PRE-SIMULATION QUIZ

Knowledge Assessment

Question 1: What is the primary purpose of Quantum Key Distribution (QKD)?

A) To encrypt data faster than classical methods

B) To distribute cryptographic keys securely between two parties

C) To detect quantum particles in real-time

D) To replace all classical encryption methods

Correct Answer: B

Question 2: What is the BB84 protocol?

A) A quantum computing algorithm for factoring large numbers

B) A quantum key distribution protocol using photon polarization

C) A method to amplify quantum signals

D) A classical encryption technique used in 1984

Correct Answer: B

Question 3: In the BB84 protocol, what are the two types of bases used?

A) Linear and Angular bases

B) Rectilinear and Diagonal bases

C) Horizontal and Vertical bases

D) Real and Imaginary bases

Correct Answer: B

Question 4: How does an eavesdropper affect quantum key distribution?

A) They cannot be detected as quantum states remain unchanged

B) Their measurement collapses the quantum state, introducing detectable errors

C) They only affect the key 50% of the time

D) Eavesdroppers are impossible to identify in QKD systems

Correct Answer: B

Question 5: What is the quantum uncertainty principle's relevance to QKD security?

A) It proves that faster-than-light communication is possible

B) It guarantees that measuring a quantum state changes it, enabling eavesdropper detection

C) It limits the number of bits that can be transmitted

D) It has no practical application in cryptography

Correct Answer: B

POST-SIMULATION QUIZ

Application & Understanding

Question 1: After running the simulator, which basis mismatch rate did you observe?

A) 0% - all bases match perfectly

B) Approximately 25% - matching occurs when both use the same basis

C) Approximately 50% - bases mismatch randomly

D) 100% - bases never match

Correct Answer: C

Question 2: In the simulator, how does the presence of an eavesdropper affect the quantum bit error rate (QBER)?

A) QBER remains at 0% regardless of eavesdropping

B) QBER increases to approximately 50% when Eve eavesdrops

C) QBER increases to approximately 25% when Eve eavesdrops

D) QBER cannot be measured in simulations

Correct Answer: C

Question 3: What was the final sifted key length after eliminating basis mismatches in your simulation?

A) The same as the original quantum bits sent

B) Approximately 25% of the original quantum bits

C) Approximately 50% of the original quantum bits

D) Zero - all bits are discarded

Correct Answer: C

Question 4: How does the simulator demonstrate the advantage of QKD over classical key exchange?

A) It transmits keys faster than light

B) It allows eavesdropper detection while classical methods do not

C) It uses fewer computational resources

D) It requires no quantum hardware

Correct Answer: B

Question 5: Based on the simulator's noise/error analysis, what would be a typical action if you detected unusual QBER levels?

A) Accept the key without further verification

B) Abort the key exchange and investigate for potential eavesdropping

C) Repeat the transmission to average out errors

D) Use the noisy key anyway for encryption

Correct Answer: B